

CAPSTONE PROJECT REPORT

Problem Statement

- A data-driven firm needs to help book publishers and online retailers make strategic decisions based on real catalogue data.
- The organization wants to understand the following:
- Which genres and price segments tend to produce the highest reader ratings?
- What are the key pricing, availability, and genre patterns across the full book catalogue?
- How can these insights be communicated through an interactive, stakeholder-ready dashboard?

OBJECTIVES

- Scrape 1,000+ book records from books.toscrape.com, capturing 10 meaningful fields per record
- Clean and preprocess the raw dataset, handling missing values and engineering new features
- Conduct Exploratory Data Analysis (EDA) to uncover statistical and visual insights
- Build and evaluate multiple classification models to predict high-rated books
- Deliver an interactive Plotly Dash dashboard communicating key insights and model outcomes

Task 1: Web Scraping

- Data was collected from books.toscrape.com.
- The scraper followed a two-level approach:
 - Level 1 — Catalogue pages: Iterating through all 50 paginated catalogue pages to collect book listings.
 - Level 2 — Detail pages: For each book, visiting its individual page to collect genre.

Task 2: Data Cleaning & Preprocessing

- Duplicate removal: Dropped duplicate rows on the 'title' column.
- No duplicates were found in the 1,000-record dataset.
- Availability standardisation: Created a binary in_stock column (1/0) using `str.contains('In stock')`.
- high_rated: Binary target variable — 1 if rating ≥ 4 , else 0 — used as the ML classification target.
- title_word_count: Number of words in the book title.
- genre_encoded: Integer encoding of genre using LabelEncoder for use in ML models.

Task 3: Exploratory Data Analysis & Visualization

- Rating Distribution as bar chart shows a near-uniform spread across ratings 1–5.
- Price Distribution as histogram reveals a right-leaning cluster in the £20–45 price range.
- The pie chart shows 98.7% of book are in-stock.
- Overlapping distributions suggest weak price-rating correlation.
- There are 1000 total books with average price 35.07 and average ratings as 3.00 stars.
- Number of distinct genres are 50

Task 4: Model Building & Evaluation

- The predictive modelling task was to classify books as high_rated with a condition if rating ≥ 4 stars then 1, otherwise 0.
- The dataset was split 80/20 into training and test sets with stratified sampling to preserve class balance.
- There were these 7 features which were used as model inputs and are price_gbp, num_in_stock, num_reviews, desc_length, title_word_count, genre_encoded, in_stock.
- The models trained are Logistic Regression, Random Forest Classifier, Gradient Boosting Classifier.
- Random Forest was selected as the best-performing model based on highest ROC-AUC(0.668) and CV accuracy 63.8%.

Task 5: Interactive Dashboard

- An interactive analytics dashboard was built using Plotly Dash with the DARKLY Bootstrap theme.
- The dashboard was designed to run in Google Colab using `app.run` and is extendable for local or cloud deployment.
- Few of the dashboard components are Header, KPI Cards, Genre Filter (Dropdown), Rating Slider, Rating Distribution Bar Chart, Price Histogram, Genre Pie Chart, Price by Rating Boxplot, Average Price by Genre Bar Chart, Feature Importance Chart.
- All the charts update dynamically via Dash callbacks using Input/Output bindings on the genre dropdown and rating range slider.

CONCLUSIONS

- The capstone project successfully demonstrated a complete, end-to-end data science workflow applied to a real-world web scraping use case.
- The key findings include: the near-uniform distribution of ratings across the catalogue, the dominance of sequential art, the strong role of price and description length as predictors of high ratings.
- The Plotly Dash dashboard provides a business-ready interface for exploring the data interactively.
- Future work should focus on incorporating semantic NLP features from description text, sourcing real-world review data from platform such as Goodreads.

Link for the video:

<https://www.loom.com/share/91457062e0944c689c1f7bc9ca55c5be>